

PROJECT DOCUMENTATION REPORT

HR Attrition Analytics Dashboard Solution

Program: Digital Egypt Pioneers Initiative

Ministry: Ministry of Communications and Information Technology

Development Tool: Microsoft Power BI Desktop

Classification: Confidential - Internal Use Only

1. Project Planning & Management

1.1 Project Proposal

Overview: This project introduces an end-to-end workforce intelligence solution built to analyze a 10-year historical cohort of 1,470 employees across an enterprise. The system transitions HR operations from traditional static reporting into a proactive retention ecosystem.

Objectives: The central goal is to isolate and quantify the structural operational, demographic, and financial drivers of workforce attrition. By exposing 'flight risks' early, the system provides actionable data to mitigate an annual replacement exposure of \$40.156M.

Scope: The workspace covers historical data from FY 2012 to FY 2022 across three operational business units: Technology, Sales, and Human Resources. It uncovers deep-seated structural retention issues like the 'Onboarding Trap' and overtime burnout cycles.

1.2 Project Plan & Timeline

The development lifecycle followed a rigorous multi-stage execution layout:

- **Data Exploration & Problem Formulation:** Initial profiling of unstructured arrays to outline high-level turnover exposure.
- **Data Prep & ETL Pipeline Engineering:** Extraction, validation, data scrubbing, and conditional grouping within Power Query.
- **Data Modeling:** Formulation of an optimized Star Schema linking transaction and dimension tables.
- **Dashboard Canvas Architecture:** Engineering interactive UI layouts, navigation elements, and cross-filtering configurations.
- **Strategic Reporting:** Financial impact forecasting, operational insights compilation, and policy formulation.

1.3 Task Assignment & Roles

The dashboard solution was built collaboratively by a dedicated technical project team under expert academic supervision:

- **Project Core Team:** Donya Tawfik, Maher Gamal, Mohammed Abdel Hady, Mustafa Zaky, Rahma Yosry, Shymaa Mohammed
- **Academic Supervisor:** Dr. Maged Madgy
- **Functional Role Split:** Data Engineers handled ETL and row validation; BI Analysts authored complex DAX calculations; UI/UX Designers optimized canvas layouts.

1.4 Risk Assessment & Mitigation Plan

Risk 1: Statistical Artifacts in Micro-Cohorts: In early tracking years (FY 2012–2019), new joiner counts were very small, leading to distortions like an apparent 100% new-hire attrition rate if a single person left. *Mitigation:* Visual indicators were added to notify stakeholders of low sample sizes, ensuring context is clear during reviews.

Risk 2: Sensitive Data Exposure: The database contains confidential salary, satisfaction scores, and manager evaluations. *Mitigation:* The database and reports are classified as 'Confidential - Internal Use Only,' with access restricted to authorized workspace roles.

1.5 Key Performance Indicators (KPIs)

The solution monitors seven foundational baseline metrics to gauge organizational health:

- Total Headcount: 1,470 employees across all units.
- Active Workforce: 1,233 current staff members (retaining 83.9% of the historical cohort).
- Exited Headcount: 237 departed employees.
- Headline Attrition Rate: 16.12% (significantly above the healthy industry benchmark of 10–12%).
- Estimated Financial Exposure: \$40.156M in replacement costs (based on 50–200% of annual salary per role).
- Compensation Disparity: Average active salary sits at \$118.9K versus \$82.3K for departed employees (a 31% gap).
- Overtime Retention Risk: A steep 30.53% attrition rate for employees working overtime hours.

2. Literature Review & Evaluation Framework

The baseline attrition rate of 16.12% was evaluated against international human capital indexes, which typically place an optimal corporate turnover benchmark at 10–12%. This confirms a structural retention gap, particularly among early-career and specialized technical roles.

Instead of relying on static spreadsheets, this project uses an interactive Power BI framework that maps operational variables—like overtime hours and business travel frequency—directly to their financial impacts, building a strong business case for policy updates.

Future iterations can expand on this descriptive and diagnostic model by integrating predictive machine learning models (such as logistic regression or random forest models) to flag specific high-risk individuals during annual review cycles.

3. Requirements Gathering

3.1 Stakeholder Analysis

- HR Executive Leadership: Requires high-level organizational KPIs, budget impact summaries, and cross-department risk metrics to plan corporate retention initiatives.
- Business Unit Managers (Tech, Sales, HR): Require granular, role-level metrics (e.g., Sales Representative vs. Software Engineer turnover) to manage day-to-day workloads and counter burnout.
- Talent Acquisition & Onboarding Teams: Require visibility into the 'Onboarding Trap' window (0–2 years tenure) to refine orientation schedules and mentorship structures.

3.2 User Stories

- Executive View: As a Chief Human Resources Officer, I want to see the total financial exposure of attrition so that I can secure budget approvals for targeted salary adjustments.
- Operational View: As a Sales Director, I want to filter attrition metrics by Overtime and Marital Status to proactively reach out to high-risk individuals before they experience burnout.
- Onboarding View: As an Orientation Coordinator, I want to see how manager experience impacts new-hire retention so that I can assign new team members to well-suited, experienced guides.

3.3 Functional & Non-Functional Requirements

Functional Requirements: Dynamic slicing across dimensions (Department, Role, Gender, Age, Overtime); clear breakdown of metrics into dedicated analysis tabs; automated projection of turnover costs based on role profiles.

Non-Functional Requirements: Fast performance with visual loading under 2 seconds during cross-filtering; high data security to protect individual review data; consistent layout and clear typography across the entire canvas.

4. System Analysis & Design

4.1 Problem Statement & Objectives

The organization faces a critical attrition rate of 16.12%, which disproportionately impacts new joiners (0–2 years tenure) and employees working high amounts of overtime. This continuous turnover creates an annual replacement cost exposure of \$40.156M, strains the long-term talent pipeline, and leaves HR stuck in a reactive cycle where high-performing but dissatisfied employees are lost unexpectedly. The goal of this project is to build an analytical framework that surfaces these hidden patterns, providing leadership with actionable, data-driven strategies to improve compensation equity and workload balance.

4.2 Software Architecture

The solution uses the standard Microsoft Power BI end-to-end business intelligence pipeline:

- Data Source Tier: Multi-source flat files containing historical employee profiles, review data, and lookup metrics.
- Data Transformation Tier: Power Query ETL engine handles column formatting, date parsing, and conditional grouping (such as generating age and distance bands).
- Analytical Database Tier: The VertiPaq in-memory columnar engine manages the compressed data model and handles quick calculation of DAX measures.
- Presentation Tier: An interactive Power BI Desktop multi-page report canvas, ready for deployment to the cloud-based Power BI Service.

4.3 Database Design & Data Modeling (Star Schema)

Fact Tables: PerformanceRating tracks transactional evaluation metrics (6,709 total review instances) including EnvironmentSatisfaction, JobSatisfaction, RelationshipSatisfaction, SelfRating, and ManagerRating.

Dimension Tables: Employee tracks central profiles for 1,470 individuals (demographics, job roles, departments, state mappings, overtime status); calendar_table drives time-intelligence calculations; EducationLevel, SatisfiedLevel, and RatingLevel provide text descriptions for code references.

Relational Rules: A 1-to-many (1:*) relationship connects Employee to PerformanceRating on EmployeeID, with filters flowing smoothly down from the Employee profiles to their historical reviews. Lookup tables connect via 1-to-many configurations to translate numerical IDs into clear text labels.

5. Implementation (Source Code & Execution)

5.1 Key DAX Measures

```
// 1. Total Headcount Summary Measure
Total_Employees = COUNT(Employee[EmployeeID])

// 2. Active Employee Count
Active_Employees = CALCULATE(COUNT(Employee[EmployeeID]), Employee[Attrition] = "No")

// 3. Exited Employee Count
Exited_Employees = CALCULATE(COUNT(Employee[EmployeeID]), Employee[Attrition] = "Yes")

// 4. Core Headline Attrition Rate
Attrition_Rate = DIVIDE([Exited_Employees], [Total_Employees], 0)

// 5. Average Tenure restricted to active employees
Average_Tenure_Active = CALCULATE(AVERAGE(Employee[YearsAtCompany]), Employee[Attrition] = "No")

// 6. New Joiner Risk Tracking (Tenure <= 2 Years)
New_Joiner_Attrition = CALCULATE([Attrition_Rate], Employee[YearsAtCompany] <= 2)

// 7. Overtime Penetration Metric
Overtime_Percentage = DIVIDE(CALCULATE(COUNT(Employee[EmployeeID]), Employee[OverTime] = "Yes"), COUNT(Employee[EmployeeID]))
```

5.2 Version Control & Collaboration

Project files—including the source datasets, the unified .pbix database file, and this detailed documentation report—are managed within a git-controlled GitHub repository. The team followed a clean feature-branching workflow, where all updates were thoroughly reviewed via Pull Requests before being merged into the main deployment branch.

6. Testing & Quality Assurance

Test Case 1: Headcount Validation: Verify total rows in the Employee table using the Total_Employees DAX measure. Expected: 1,470 | Actual: 1,470 (Pass).

Test Case 2: Attrition Math Alignment: Cross-check that the sum of Active Employees (1,233) and Exited Employees (237) matches the total headcount. Expected: 1,470 | Actual: 1,470 (Pass).

Test Case 3: Performance Calibration Audit: Audit the entire 6,709-row review dataset for instances where a manager's rating exceeded the employee's self-rating. Expected: A natural variance. Actual Discovery: Exactly 0 reviews showed ManagerRating > SelfRating. This unexpected finding exposed a major calibration flaw in the review process, where managers never rated an employee higher than the employee's own self-assessment.

7. Technical Insights & Strategic Recommendations

7.1 Deep-Dive Technical Insights

- **Overtime is the Strongest Predictor:** Employees working overtime face a 30.53% attrition rate, nearly three times higher than the 10.44% rate for those who do not. When combined with being single and traveling frequently, attrition shoots up to a critical 52.4%.
- **Structural Crises in Commercial Roles:** Sales Representatives (39.76%) and Recruiters (37.50%) suffer from extreme turnover, meaning the entire sales force is effectively replaced every 2.5 years, creating continuous and costly hiring cycles.
- **The Stock Option Retention Gap:** Employees holding Level 0 stock options have an attrition rate of 24.4%. Providing even a baseline incentive drop (moving to Level 1) triggers a sharp 14.7 percentage point decrease in turnover, dropping attrition to just 9.4%.
- **Performance Review Calibration Flaws:** On average, employees rate themselves 0.51 points higher than their managers do. Combined with the zero-case manager over-rating anomaly, the data reveals a review process that lacks credibility and may be driving frustration among top talent.
- **The Manager Experience Variable:** New hires assigned to managers who are in their first 1-2 years of leadership face the highest turnover risk across all departments. New hire turnover drops significantly once a manager has crossed the 3-year experience mark.

7.2 Strategic Retention Intervention Matrix

INTERVENTION PROGRAM	TARGET AUDIENCE	EXPECTED BUSINESS IMPACT
1. Overtime Policy Reform - 8-hour weekly cap - Automatic HRIS flags	Sales & Technology Units experiencing extreme workload pressure.	Reduces OT attrition down to 18-20%, retaining 30-40 staff members annually.
2. Targeted Equity Shift - Move from Level 0 to Level 1 options	Sales Reps, Recruiters, and Data Scientists on base pay.	Closes the Level 0 equity gap and drops specialized role attrition by 35-50%.
3. Structured Onboarding - 3-year tenure managers - 30/60/90 day reviews	All new joiners within their first 24 months of service.	Stabilizes early-tenure teams; saves an estimated \$10M-\$16M in hiring costs.

8. User Manual & Dashboard Navigation Guide

- **Interacting with Visuals:** Every chart on the Power BI dashboard canvas is fully interactive. Clicking on a specific department bar (e.g., Sales) automatically filters the entire page to show metrics, age distributions, and attrition rates unique to that business unit.
- **Using Filter Slicers:** Use the top drop-down filter panel to look at specific risk groups. For example, selecting Marital Status: Single, Overtime: Yes, and Business Travel: Frequent Travel isolates the company's highest-risk employee segment for immediate attention.
- **Analyzing Trends over Time:** Use the line charts on the Attrition Trends sub-page to look at year-over-year changes, helping trace historic spikes (like the anomalous trend in 2016) back to restructuring events or changing market conditions.